

Chengqi Xue

Tel: +44-7493108426 Email: chengqi.xue2002@gmail.com
Robotics & Machine Learning & Computer Vision & Automation



Education

MSc. Robotics | King's College London, UK

09/2025-09/2026

Core Modules: *Robot Dynamics and Control, Robot Kinematics and Motion Planning, Sensing and Perception, Machine Learning, Intelligence and Autonomy*

BEng. Automation | Yangtze University, China

09/2020-07/2024

GPA: 3.71/5.0 Ranking: Top 7% National Scholarship (4%)

Core Modules: *C Language Programming, Computer Control Technology, Analog Electronic Technology, Digital Electronic Technology, Principle of Automatic Control, Motion Control System, Machine Vision*

Research & Project Experience

Assistive Robotics for Chemotherapy Patients | KCL

09/2025-04/2026

- Built a TIAGo robot simulation in Webots integrated with ROS 2, RViz, SLAM, and Navigation2 for autonomous indoor assistive robot research
- Developed a full SLAM and navigation pipeline enabling environment mapping, localization, global/local path planning, and obstacle avoidance
- Designed a perception module combining speech recognition and waving detection to identify patients needing assistance and supply navigation targets
- Implemented AMCL-based pose estimation to improve reliable navigation in structured indoor environments

Vision-Tactile-Language-Action Learning for Robotic Cloth Sorting | KCL

09/2025-04/2026

- Proposed a VTLA framework extending Vision-Language-Action (VLA) models with a tactile modality for contact-rich manipulation tasks
- Designed a multi-modal tokenization pipeline fusing vision, tactile, and language inputs, fine-tuning Qwen2.5-VL-7B with QLoRA on 3,937 fabric-comparison conversations
- Developed a tactile-image (GelSight sensing) force estimator for finray grippers to improve grasp stability during cloth sorting
- Developing a $\pi 0.5$ VLA-based language-conditioned action policy for Franka cloth manipulation

Research Assistant

03/2022-07/2025

Internal Thread Defect Detection System Based on Multi-vision and Deep Learning Technology, Yangtze University

- **Published 2 SCI Q1 journal papers, 2 EI international conference papers, and 2 patents**
- Designed a computer-controlled multi-camera system equipped with a fish-eye lens to monitor the internal thread area comprehensively;
- Developed a cylindrical-model-based image composition technique to clearly display thread features from curved surfaces;
- Proposed a GAN-based data augmentation pipeline to generate high-quality, diverse defect samples and expand the training dataset;
- Trained and optimized YOLO-based detection models, achieving high-precision internal thread defect recognition;

Research Assistant

03/2021- 07/2025

Research on Optimization of Engineering Material Properties Based on Machine Learning, Yangtze University

- **Published 2 SCI Q1 journal papers**
- Collected and preprocessed data on GFRP-filled concrete columns and tempered laminated glass, applying normalization and genetic-algorithm-based augmentation;
- Built machine learning models for data-driven modelling and performance optimization of composite structures
- Analyzed model predictions and proposed optimal design solutions for material performance improvement;

- **Filed 2 utility patents:**
- Designed and realized industrial monitoring and warning systems;
- Developed image acquisition, distortion correction, and enhancement algorithms for industrial monitoring cameras
- Implemented real-time image processing and data analysis algorithms to improve detection and response efficiency
- Integrated and debugged subsystems to optimize end-to-end monitoring and early warning performance

Publications

Journal Papers (SCI Q1)

1. **Chengqi Xue** (Co-first Author), T. Deng, G. Zhang, “Data Modeling Analysis of GFRP Tubular Filled Concrete Column Based on Small Sample Deep Meta Learning Method,” *PLOS ONE* <https://doi.org/10.1371/journal.pone.0305038>
2. **Chengqi Xue** (Co-first Author), X. Dou, G. Zhang, Z. Jiang, “Internal Thread Defect Detection System Based on Multi-vision,” *PLOS ONE*. <https://doi.org/10.1371/journal.pone.0304224>
3. **Chengqi Xue** (Co-first Author), P. Shu, G. Zhang, T. Deng, “Study on Design Optimization of GFRP Tubular Column Composite Structure Based on Machine Learning Method”. *PLOS ONE*. <https://doi.org/10.1371/journal.pone.0301865>
4. Z. Jiang, X. Dou, X. Liu, **Chengqi Xue**, et al. “Internal Thread Defect Generation Algorithm and Detection System Based on Generative Adversarial Networks and You Only Look Once”. *Sensors* 2024, 24(17),5636. <https://doi.org/10.3390/s24175636>

Conference Papers (EI / SPIE)

1. H. Yuan, X. Dou, **Chengqi Xue (Corresponding Author)**, et al. “Review of image calibration and distortion correction based on internal threads” *Proc. SPIE* (AASIP 2024) <https://doi.org/10.1117/12.3045519>
2. Z. Xu, **Chengqi Xue (Corresponding Author)**, X. Dou, et al., “A Review of Image Stitching Algorithms Based on Internal Thread”. *Proc. SPIE* (RSTIP 2024). <http://dx.doi.org/10.1117/12.3065383>

Under Review

1. **Chengqi Xue**, X. Zhao, H. Yuan , Y. Zhang, “Modeling Analysis of Toughened Sandwich Glass Based on Machine Learning Methods”. *PLoS ONE*. Under Review.

Patents

First Inventor, **A Wellhead Monitoring Device Based on Machine Vision**, Patent No.: ZL 202323487863.X

First Inventor, **A Wind Event Early Warning System**, Patent No.: ZL 2023 2 3530543.8

Third Inventor, **A Device for Inspecting External Threads of Tubular Tools**, Patent No.: 202411050637.1

Third Inventor, **Internal Thread Defect Detection System**, Patent No.: 202411066290.X

Honors & Awards

Best Researcher Award at the International Research Awards on Sensing Technology	09/2024
National Scholarship, Yangtze University	12/2023
Third prize, (Provincial level), National Undergraduate Electronic Design Contest	10/2023
First Prize, (National level), College students AI Technology Competition Knowledge Contest	09/2023
Second Prize, College Students “Internet” Innovation and Entrepreneurship Competition	07/2022

Skills

Proficient in using Python, MATLAB, Linux, RViz, ROS 2, SLAM, Navigation2 Gazebo, and etc.